Ch 7: Consumers, Producers, and the Efficiency of Markets

Consumer surplus, producer surplus, and what a market does well

Onur Altındağ

EC 111 Principles of Microeconomics, Bentley University, Fall 2026

From who pays to how much it costs

Chapter 6

Policy moves P and Q

We said who gains and who loses

Chapter 7

How much, in dollars?

Is the market outcome good for society?

Welfare economics

How the allocation of resources affects people's economic well-being

Two measures:

consumer surplus — what buyers gain

producer surplus — what sellers gain

Willingness to pay

Four fans, one Red Sox playoff ticket each

Fan	Willingness to pay
Ana	$250
Ben	$200
Cleo	$150
Dev	$100

Willingness to pay

The most a buyer would pay for a good: what it is worth to her

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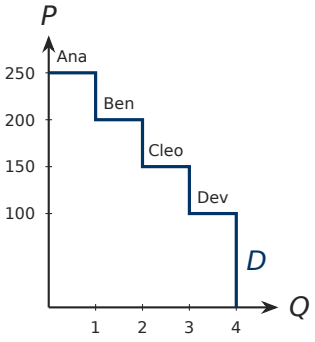
The most a buyer would pay for a good: what it is worth to her

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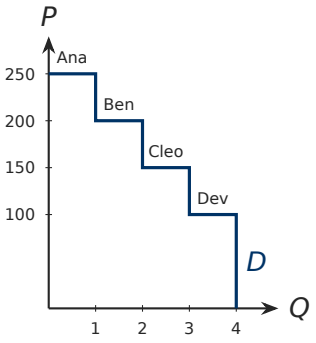
She values it at $250 and pays $180: she gains $70.

Willingness to pay traces out the demand curve



Price	Tickets demanded
above $250	0
$200–$250	1
$150–$200	2
$100–$150	3
$100 or less	4

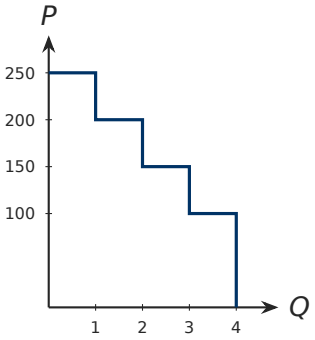
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The height of the demand curve at each Q is the willingness to pay of the marginal buyer: the one who would leave first if the price rose.

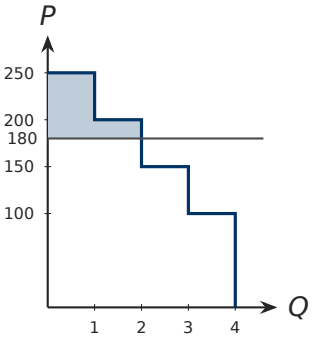
Consumer surplus



Consumer surplus

Willingness to pay minus the price paid, summed over buyers

Consumer surplus



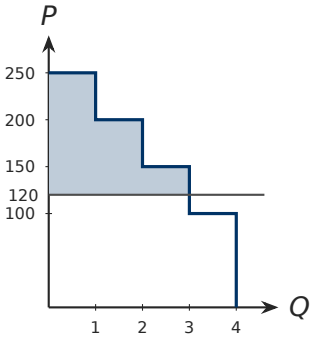
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At \$180: Ana and Ben buy

$$\text{CS} = 70 + 20 = \$90$$

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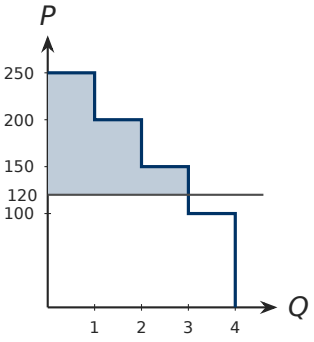
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At \$120: Cleo buys too

$$\text{CS} = 130 + 80 + 30 = \$240$$

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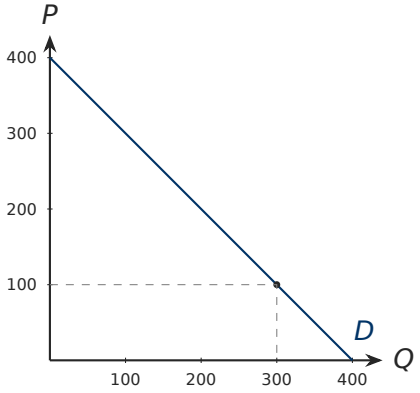
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CS is the area below the demand curve and above the price.

With many buyers: a triangle

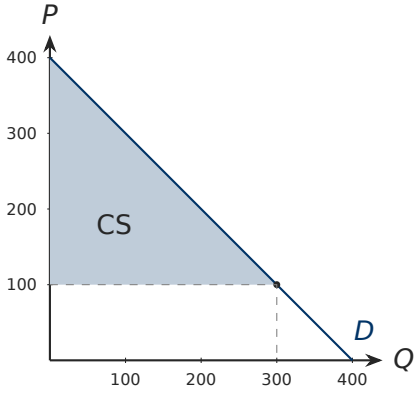


The market from Ch 6:

$$Q_d = 400 - P, \quad P^* = 100, \quad Q^* = 300$$

Demand reaches $Q = 0$ at $P = 400$:
no buyer values the good above \$400

With many buyers: a triangle



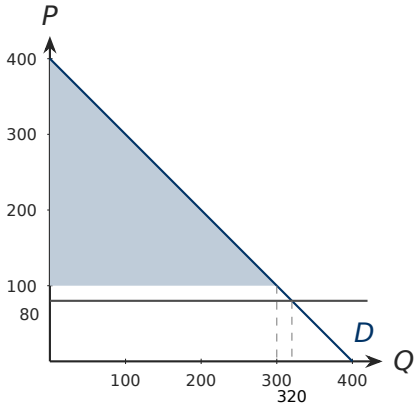
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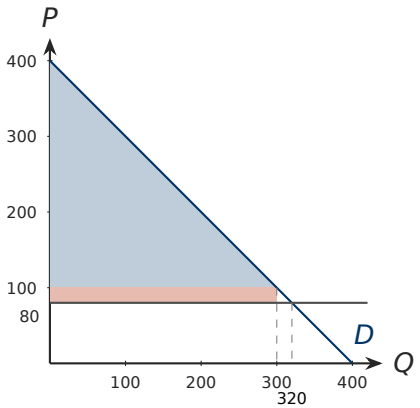
$$\text{CS} = \tfrac{1}{2} \times \text{base} \times \text{height}$$
$$= \tfrac{1}{2} \times 300 \times (400 - 100)$$
$$= \$45{,}000$$

The price falls: who gains?



Supply grows and the price falls from $100 to $80

The price falls: who gains?

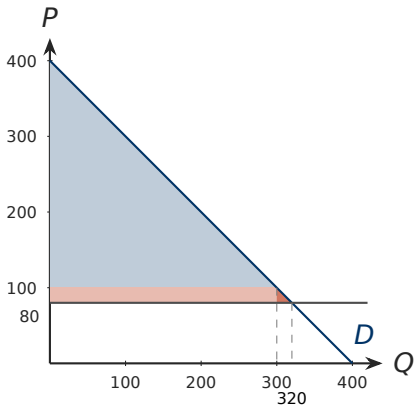


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Existing buyers pay \$20 less on 300 units

$300 \times 20 = \$6{,}000$

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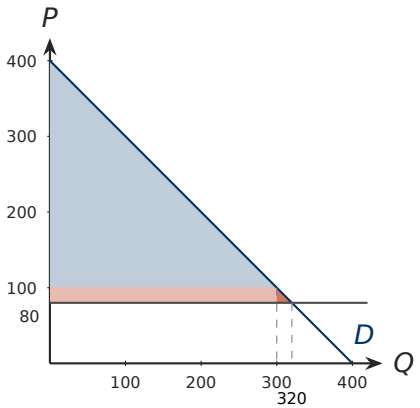
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$\frac{1}{2} \times 20 \times 20 = \$200$

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CS rises from \$45,000 to

$\frac{1}{2} \times 320 \times 320 = \$51{,}200$

What consumer surplus measures

The benefit buyers get from taking part in the market, as they see it

A good measure of well-being when buyers' preferences are worth respecting

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Should a policymaker count the consumer surplus of a heroin buyer?

Most economists say yes for ordinary goods: people are the best judges of what they want.

Addiction is the usual exception: the buyer's own willingness to pay may not reflect her well-being.

Cost and willingness to sell

Four students could each tutor one hour of economics

Tutor	Cost
Eli	$20
Fay	$30
Gus	$45
Hal	$60

Cost

The value of everything a seller gives up to produce the good, including her time: the lowest price at which she will sell

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Eli tutors for $50. How much better off is he?

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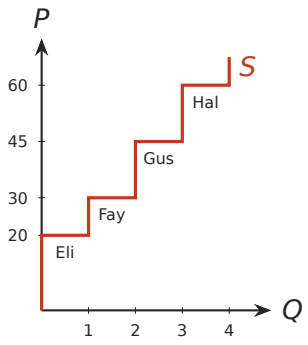
The value of everything a seller gives up to produce the good, including her time: the lowest price at which she will sell

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His cost is $20 and he is paid $50: he gains $30.

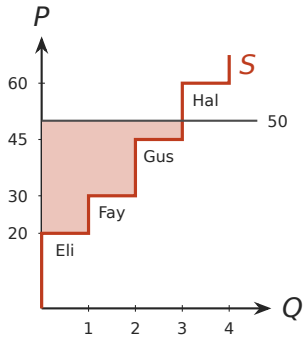
Producer surplus



Producer surplus

The price received minus the seller's cost, summed over sellers

Producer surplus



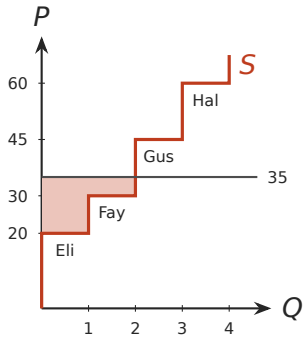
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At \$50: Eli, Fay and Gus tutor

$\text{PS} = 30 + 20 + 5 = \$55$

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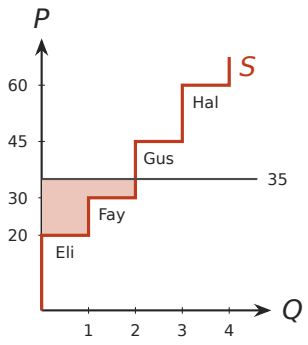
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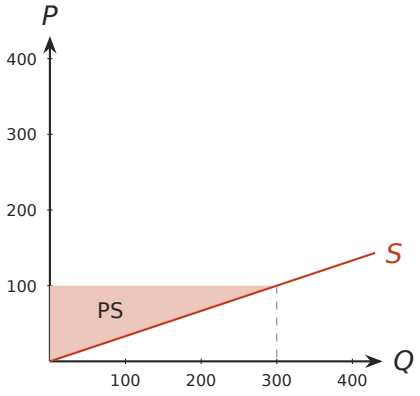
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The supply curve is the cost of the marginal seller. PS is the area above the supply curve and below the price.

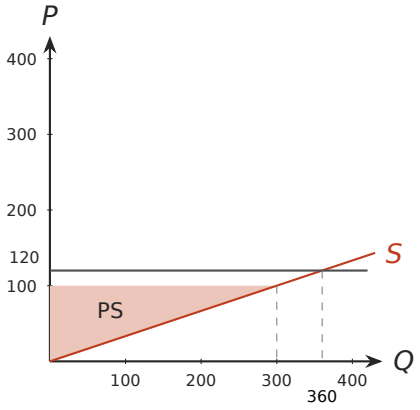
Producer surplus in the running market



$Q_s = 3P$. At \$100:

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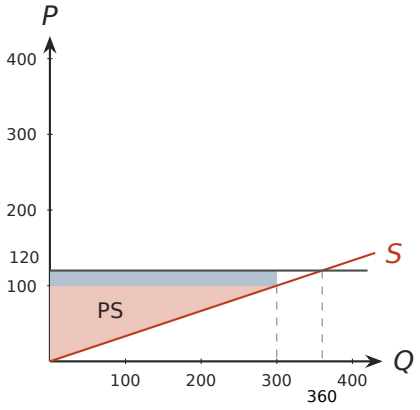


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Demand grows and the price rises to \$120

Producer surplus in the running market



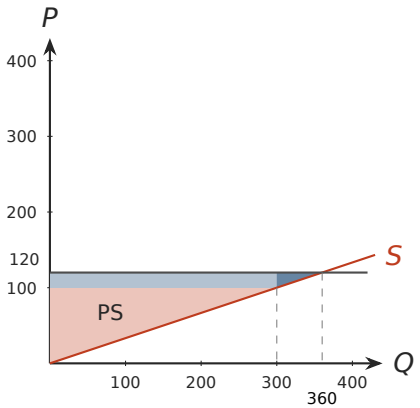
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Existing sellers get \$20 more on 300 units: \$6,000

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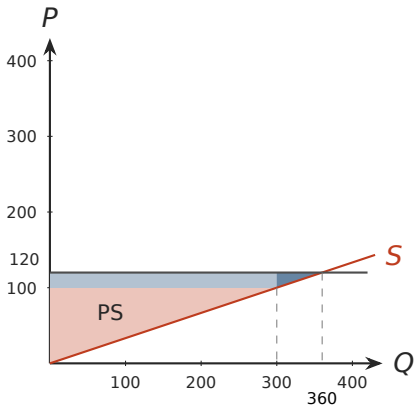
Demand grows and the price rises to \$120

Existing sellers get \$20 more on 300 units: \$6,000

New sellers: 60 more units, each costing less than \$120:

$\frac{1}{2} \times 60 \times 20 = \$600$

Producer surplus in the running market



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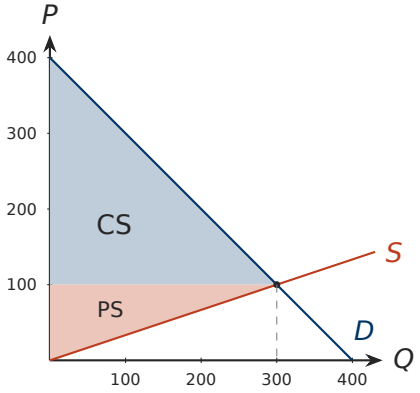
Existing sellers get \$20 more on 300 units: \$6,000

New sellers: 60 more units, each costing less than \$120:

$\frac{1}{2} \times 60 \times 20 = \$600$

PS rises to $\frac{1}{2} \times 360 \times 120 = \$21{,}600$

Total surplus



Total surplus = CS + PS

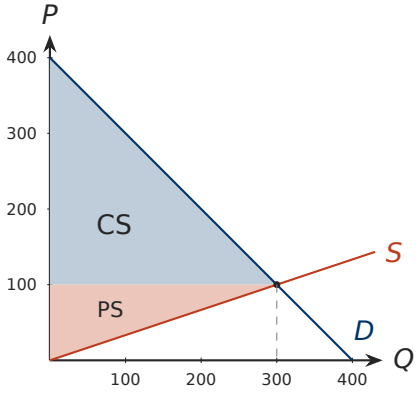
CS = value to buyers − price

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Add them; the price cancels:

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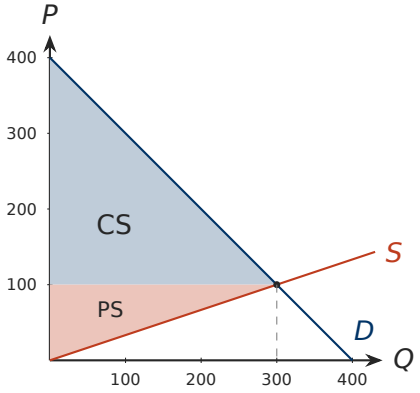
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Running market:

$45{,}000 + 15{,}000 = \$60{,}000$

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45,000 + 15,000 = $60,000

Efficiency

An allocation is efficient if it maximizes total surplus

Efficiency and equality

Efficiency: is the pie as large as possible?

Equality: is the pie divided fairly?

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Question

Ana (willingness to pay $250) buys a ticket from a seller whose cost is $140. How much total surplus does the trade create? Does it depend on the price?

Efficiency and equality

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Equality: is the pie divided fairly?

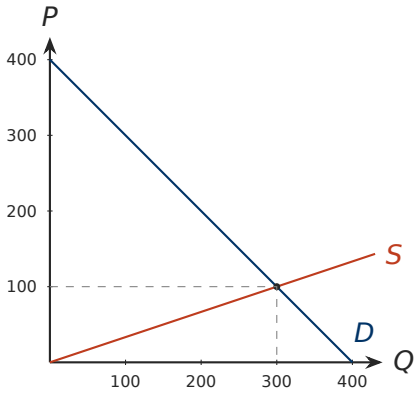
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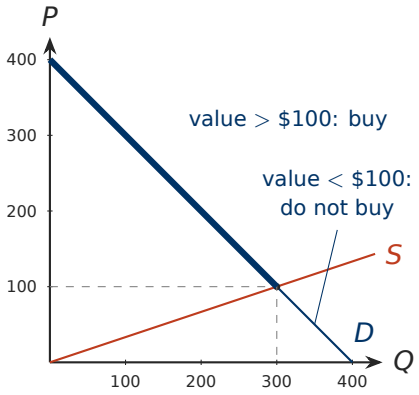
$250 - 140 = \$110$, at any price. At \$200, Ana gets 50 and the seller 60; at \$150, Ana gets 100 and the seller 10.

What the market gets right: the buyers



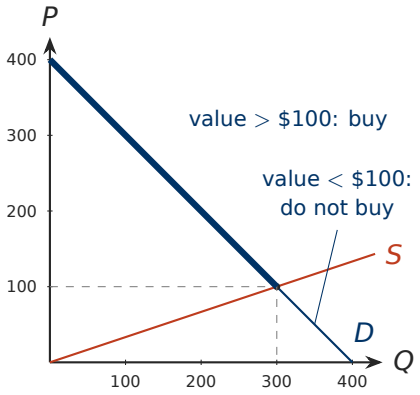
At $P = \$100$: buyers who value it above \$100 buy; the rest do not

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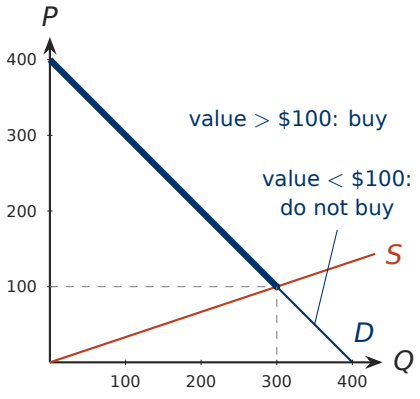


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Insight 1

A free market gives the goods to the buyers who value them most

What the market gets right: the buyers



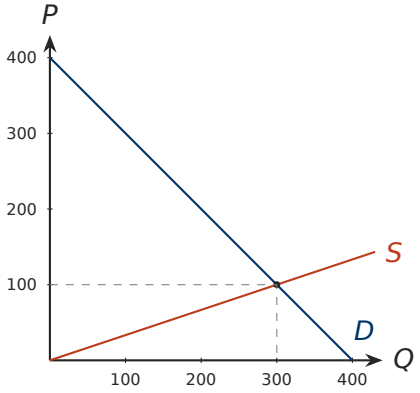
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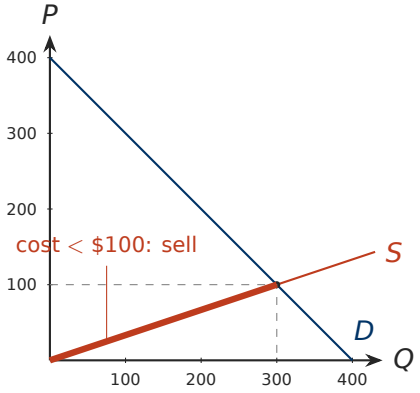
A lottery for the 300 units would give some to buyers who value them below \$100 and leave out some who value them more.

What the market gets right: the sellers



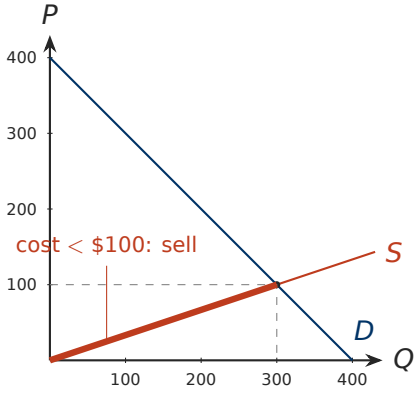
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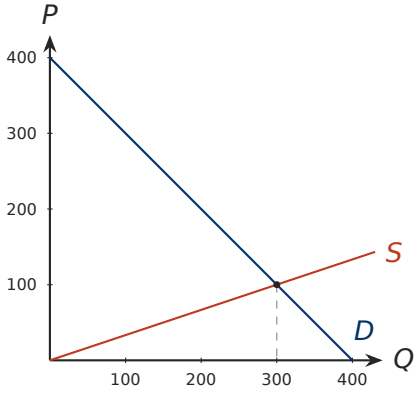


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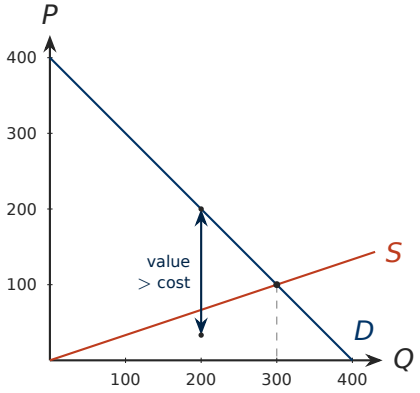
Insight 2

A free market gets the goods made by the sellers who produce them at the lowest cost

What the market gets right: the quantity

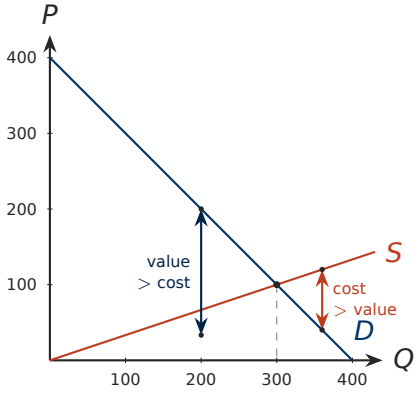


What the market gets right: the quantity



At $Q = 200$: the next unit is worth \$200 to a buyer and costs \$67 to make. Producing it adds \$133.

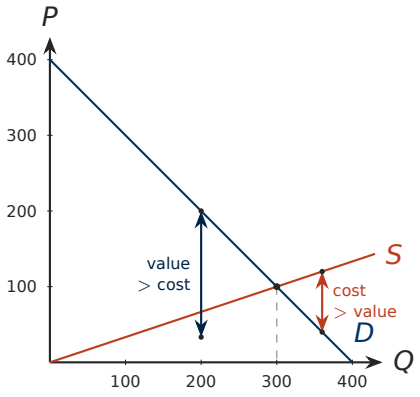
What the market gets right: the quantity



At $Q = 200$: the next unit is worth \$200 to a buyer and costs \$67 to make. Producing it adds \$133.

At $Q = 360$: the next unit is worth \$40 and costs \$120. Producing it loses \$80.

What the market gets right: the quantity



At $Q = 200$: the next unit is worth \$200 to a buyer and costs \$67 to make. Producing it adds \$133.

At $Q = 360$: the next unit is worth \$40 and costs \$120. Producing it loses \$80.

Insight 3

A free market produces the quantity that maximizes total surplus: $Q^* = 300$, where value = cost

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No one in the market is trying to maximize total surplus. Each buyer and seller looks after their own interest.

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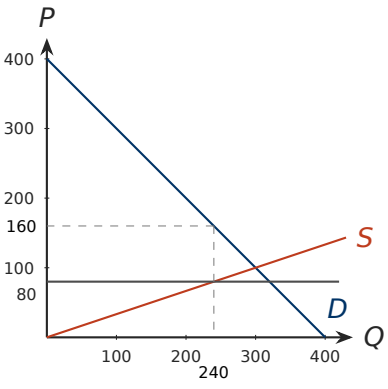
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Adam Smith, 1776

"…led by an invisible hand to promote an end which was no part of his intention."

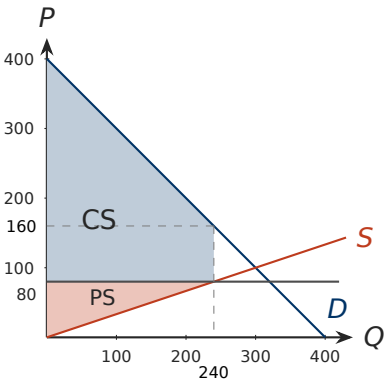
Ch 4: Hayek and the price as a signal. Ch 7 puts a dollar figure on it.

Back to Ch 6: the $80 ceiling



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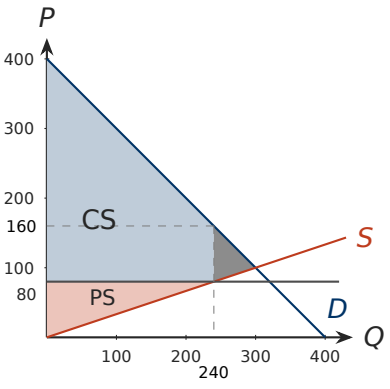
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$$\text{CS} = 240 \times 80 + \tfrac{1}{2} \times 240 \times 240 = \$48{,}000$$

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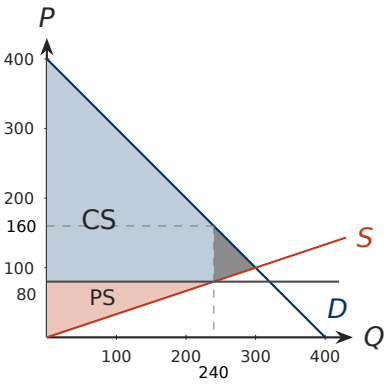
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Buyers gain \$3,000, sellers lose \$5,400. Rationing by lines or luck loses more.

When the market does not get it right

The three insights assume perfectly competitive markets and that the price reflects all costs and benefits

Market power

One seller (or a few) can keep the price above cost and the quantity below Q^*

Ch 16–17

Externalities

A cost or benefit falls on people outside the market: pollution, vaccination

Buyers and sellers ignore it, so Q is too high or too low

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These are market failures: cases where policy can raise total surplus.

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Against it: fairness (the poor sell, the rich buy), exploitation, and the view that some things should not be for sale. Efficiency is one goal, not the only one.

Try it

Question

$Q_d = 90 - P, \quad Q_s = 2P.$

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(c) Value $90 - 40 = \$50$, cost $40/2 = \$20$. Deadweight loss $\frac{1}{2} \times 20 \times (50 - 20) = 300$, so TS $= 2{,}400$.

Exam-style: four buyers, four sellers

Buyer	Value	Seller	Cost
A	250	E	60
B	200	F	90
C	150	G	140
D	100	H	210

Each buys or sells one unit.

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(b) What prices clear the market?
(c) Total surplus?

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(c) $(250 + 200 + 150) - (60 + 90 + 140)$ $= 600 - 290 = \$310$, whatever the price in that range.

Summary

Surplus

- Willingness to pay: the most a buyer would pay. The demand curve's height
- Consumer surplus = value − price: area under D, above P
- Cost: the least a seller would accept. The supply curve's height
- Producer surplus = price − cost: area above S, below P
- A lower price raises CS; a higher price raises PS

Efficiency

- Total surplus = value to buyers − cost to sellers
- Efficient = total surplus as large as possible; equality is a separate question
- A competitive equilibrium gives goods to the highest-value buyers, gets them made at the lowest cost, and produces Q^*
- Price controls and market failures lower total surplus; the loss is the deadweight loss (more in Ch 8)

Next: Ch 8, the costs of taxation.

MindTap homework due Sun Oct 25, 11 PM.